

6. Explain the difference between array and structures

Arrays are collection of same data type elements whereas in structures elements of different data types can also be stored together.

Example:

```
int arr[5]; //an array to stores 5 integers
struct st
{
    int arr[5];
    float b;
    char c;
    double c;
}; //a structure storing an integer array, float, character and double elements together.
```

11. Write a program that takes as input two-time instants (let say: t1(h1,m1,s1) and t2(h2,m2,s2) where h, m and s are hours, minutes and seconds resp.) and finds the difference between them (t1-t2) using the concept of structures.**INPUT:**

Enter hrs, mins, secs for t1:

03

20

50

Enter hrs, mins, secs for t2:

02

10

30

OUTPUT:

Difference= 1:10:20

```
#include <stdio.h>
typedef struct time
{
    int sec;
    int min;
    int hrs;
}time;
```

```
void difference(time, time, time *);
```

```
int main()
{
    time t1, t2, diff;
```

```
printf("Enter hrs, mins, secs for t1:\n ");

scanf("%d %d %d", &t1.hrs, &t1.min, &t1.sec);

printf("Enter hrs, mins, secs for t2:\n ");
scanf("%d %d %d", &t2.hrs, &t2.min, &t2.sec);

difference(t1,t2, &diff);
printf("Difference= %d:%d:%d\n", diff.hrs, diff.min, diff.sec);
printf("\n ----END OF PROGRAM-----\n");
return 0;
}

void difference (time start, time stop, time *diff)
{
    if(stop.sec > start.sec)
    {
        start.min--;
        start.sec = start.sec + 60;
    }

    diff->sec = start.sec - stop.sec;
    if(stop.min > start.min)
    {
        start.hrs--;
        start.min = start.min + 60;
    }

    diff->min = start.min - stop.min;
    diff->hrs = start.hrs - stop.hrs;
}
```